

HW 6.1.1: Movement along a Polygonal Path

For this assignment, we are concerned with the movement of an object along a path in the plane. We are assuming that the plane is a coordinate plane and the object starts at the point $(0,0)$. As the object moves long the path, each point on that path has two coordinates. The coordinates depend on the distance traveled along the path. Let us call this distance S , the length of the path from the origin to a point P on the path.

1. Evaluate the following

a.) What value of S yields the coordinate point $(2, 3)$?

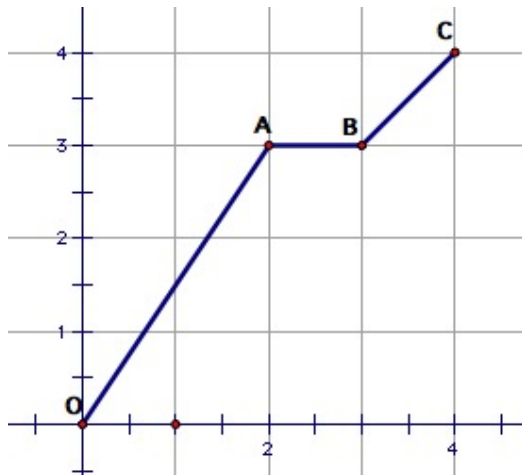
b.) What value of S yields the coordinate point $(4, 4)$?

c.) $x(\sqrt{13}) =$

d.) $y(1 + \sqrt{13}) =$

e.) $x(2) =$

f.) $y(4) =$



(Challenge) Write explicit formulas (piecewise functions) describing the path of $x(S)$ and $y(S)$.

2. Evaluate the following:

a.) What value of S yields the coordinate point $(2, 3)$?

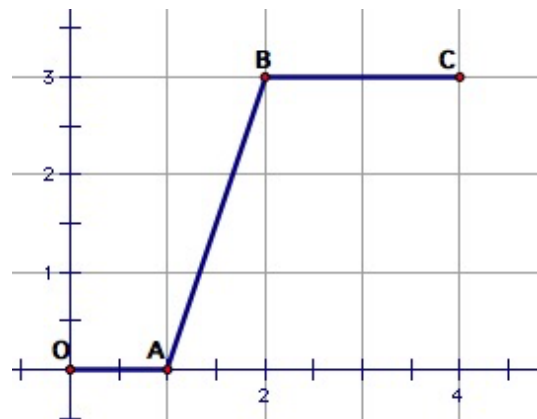
b.) What value of S yields the coordinate point $(4, 3)$?

c.) $x(1) =$

d.) $y(1 + \sqrt{10}) =$

e.) $x(2 + \sqrt{10}) =$

f.) $y(3 + \sqrt{10}) =$



(Challenge) Write explicit formulas (piecewise functions) describing the path of $x(S)$ and $y(S)$.

3. Evaluate the following:

a.) What value of S yields the coordinate point $(1, 3)$?

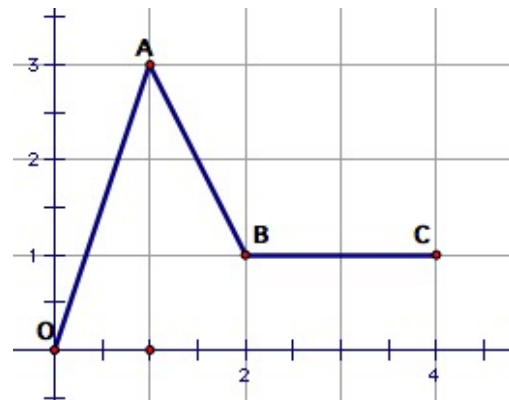
b.) What value of S yields the coordinate point $(3, 1)$?

c.) $x(\sqrt{10}) =$

d.) $y(\sqrt{10} + \sqrt{5}) =$

e.) $x(2) =$

f.) $y(2.5) =$



(Challenge) Write explicit formulas (piecewise functions) describing the path of $x(S)$ and $y(S)$.

Answers:

1. a. $s = \sqrt{13}$ b. $s = \sqrt{13} + 1 + \sqrt{2}$ c. 2 d. 3 e. $\frac{4\sqrt{13}}{13} \approx 1.109$ f. 3

Challenge:

$$x(s) = \begin{cases} \frac{2}{\sqrt{13}}s & \text{if } 0 \leq s < \sqrt{13} \\ s - \sqrt{13} + 2 & \text{if } \sqrt{13} \leq s < \sqrt{13} + 1 \\ \frac{1}{\sqrt{2}}(s - \sqrt{13} - 1) + 3 & \text{if } \sqrt{13} + 1 \leq s \leq \sqrt{13} + 1 + \sqrt{2} \end{cases}$$

$$y(s) = \begin{cases} \frac{3}{\sqrt{13}}s & \text{if } 0 \leq s < \sqrt{13} \\ 3 & \text{if } \sqrt{13} \leq s < \sqrt{13} + 1 \\ \frac{1}{\sqrt{2}}(s - \sqrt{13} - 1) + 3 & \text{if } \sqrt{13} + 1 \leq s \leq \sqrt{13} + 1 + \sqrt{2} \end{cases}$$

2. a. $s = 1 + \sqrt{10}$ b. $s = 3 + \sqrt{10}$ c. 1 d. 3 e. 3 f. 3

Challenge:

$$x(s) = \begin{cases} s & \text{if } 0 \leq s < 1 \\ \frac{s-1}{\sqrt{10}} + 1 & \text{if } 1 \leq s < \sqrt{10} + 1 \\ s - (1 + \sqrt{10}) + 2 & \text{if } \sqrt{10} + 1 \leq s \leq \sqrt{10} + 3 \end{cases}$$

$$y(s) = \begin{cases} 0 & \text{if } 0 \leq s < 1 \\ (s-1) \cdot \frac{3}{\sqrt{10}} & \text{if } 1 \leq s < \sqrt{10} + 1 \\ 3 & \text{if } \sqrt{10} + 1 \leq s \leq \sqrt{10} + 3 \end{cases}$$

3.

Challenge:

$$x(s) = \begin{cases} \frac{s}{\sqrt{10}} & \text{if } 0 \leq s < \sqrt{10} \\ \frac{(s - \sqrt{10})}{\sqrt{5}} + 1 & \text{if } \sqrt{10} \leq s < \sqrt{10} + \sqrt{5} \\ s - (\sqrt{10} + \sqrt{5}) + 2 & \text{if } \sqrt{10} + \sqrt{5} \leq s \leq \sqrt{10} + \sqrt{5} + 2 \end{cases}$$

$$y(s) = \begin{cases} s \cdot \frac{3}{\sqrt{10}} & \text{if } 0 \leq s < \sqrt{10} \\ 3 - (s - \sqrt{10}) \cdot \frac{2}{\sqrt{5}} & \text{if } \sqrt{10} \leq s < \sqrt{10} + \sqrt{5} \\ 1 & \text{if } \sqrt{10} + \sqrt{5} \leq s \leq \sqrt{10} + \sqrt{5} + 2 \end{cases}$$